Charges and Charging, and Coulomb's Law

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1 Charging Introduction

These are my notes on charges and charging for Physics!

Charges are fundamental properties for particles, just like **mass** and **spin**.

2 Fundamental Properties of Particles

There are three particles that matter for charges and charging.

Particle	Charge (in e)	Charge (in Coulombs (C))
Proton	1 e	1.6×10^{-19} C
Electron	- 1 e	-1.6×10^{-19} C
Neutron	1 e	1.6×10^{-19} C

Charges are usually sent in multiples of elementary charge, usually through electrons (because if you remove protons, the atoms themselves change).

Note: Elementary charges are elementary because they are the smallest stable charge. You can get quarks (with fractions of elementary charges), but will reform to these particles.

3 Sending and Recieving Charge

There are three methods for sending and recieving charge:

- Friction
- Conduction
- Induction

3.1 Friction

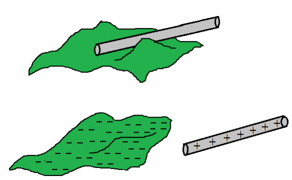

Figure 1: Example of Friction

When two objects rub with each other, the object with higher affinity will **take** electrons from the other.
Usually, if you have two of the same object rub the same way, the objects have the same charge.

3.2 Conduction

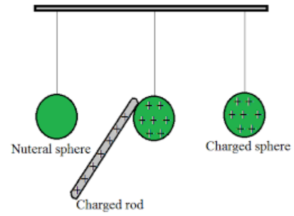

Figure 2: Example of Conduction

When two conductors (all objects are conductors in the end, if you try hard enough). If objects are identifical, their charge will even out. Elsewise, they will just equalize.

Note: These conductors equalize to minimize the system's *kinetic energy*.

3.3 Induction

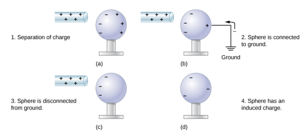

Figure 3: Example of Induction

A charge can change through proximity to an object. To equalize the charge afterwards, you can just tap the two objects together (if I remember correctly).

4 Polarized Neutral

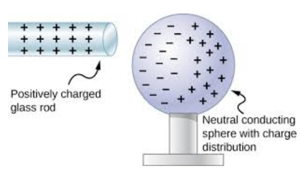

Figure 4: Example of a polarized neutral

A neutral object can be polarized through having the charged object attract the opposite charge of a neutral object, causing attraction through the distance.

5 Question Stuff

The question worksheet was *Electric Charge and Methods of Charging*, and the only weird thing was question 4. To solve, just do

$$\frac{1.5\,\mathrm{m} \times \left(6.023 \times 10^{23}\,\text{particles per m}\right)}{1 \times 10^{6}\,\text{particles per e}} \times \frac{1.6 \times 10^{-19}\,\mathrm{C}}{e} = 0.144\,\mathrm{C}$$

6.023×10^{23} is Avogadro's (Avocado's) number, which is equal to the number of particles per mole. One just has to divide by that the number of particles before one is charged, and then to convert elementary charges to coulombs.

6 Coulomb's Law

Coulomb's Law is related to how two charged particles are attacted to each other.

6.1 Coulombs's Law from Regents

$$F_e = \frac{kq_1q_2}{R^2}$$

Problems with this formula:

- k assumes a vacuum
- q_1 , q_2 are point particles
- One has to assume one object is stationary

6.2 Building from Coulomb's Law for Physics C

1. We don't want to use k anymore. Instead we use $\frac{1}{4\pi\epsilon}$ ϵ represents the permittivity of a material [1]. For vacuums, the permittivity, or ϵ_0, is equal to $8.85 \times 10^{-12} \frac{\mathrm{C}^2}{\mathrm{Nm}^2}$
2. To fix the fact that that q_1 and q_2 must be point particles, one will have to integrate over δq, or change in charge, to find the total charge, and integrate over δF. to find the force.

Putting everything together, we have the new formula:

$$\boxed{\vec{F_e} = \frac{1}{4\pi\epsilon}\frac{q_1q_2}{R^2}\hat{r}}$$

where $\hat{r}$ is vector pointing to the other charge. Remember, $\boxed{\epsilon_0 = 8.85 \times 10^{-12}\frac{\mathrm{C}^2}{\mathrm{Nm}^2}}$

[1]the ablity of a particle to hold enerty at a specific field without breaking down